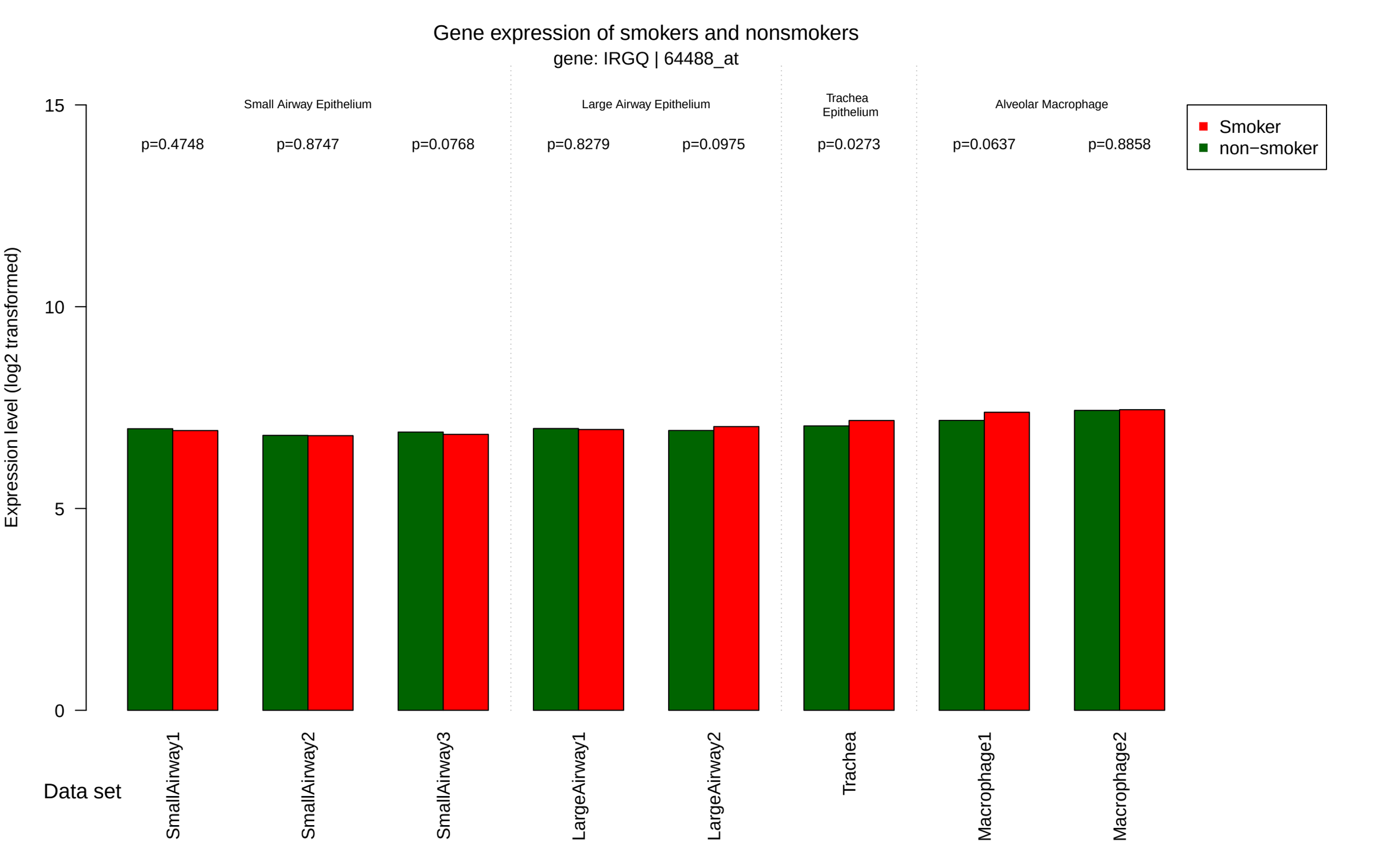
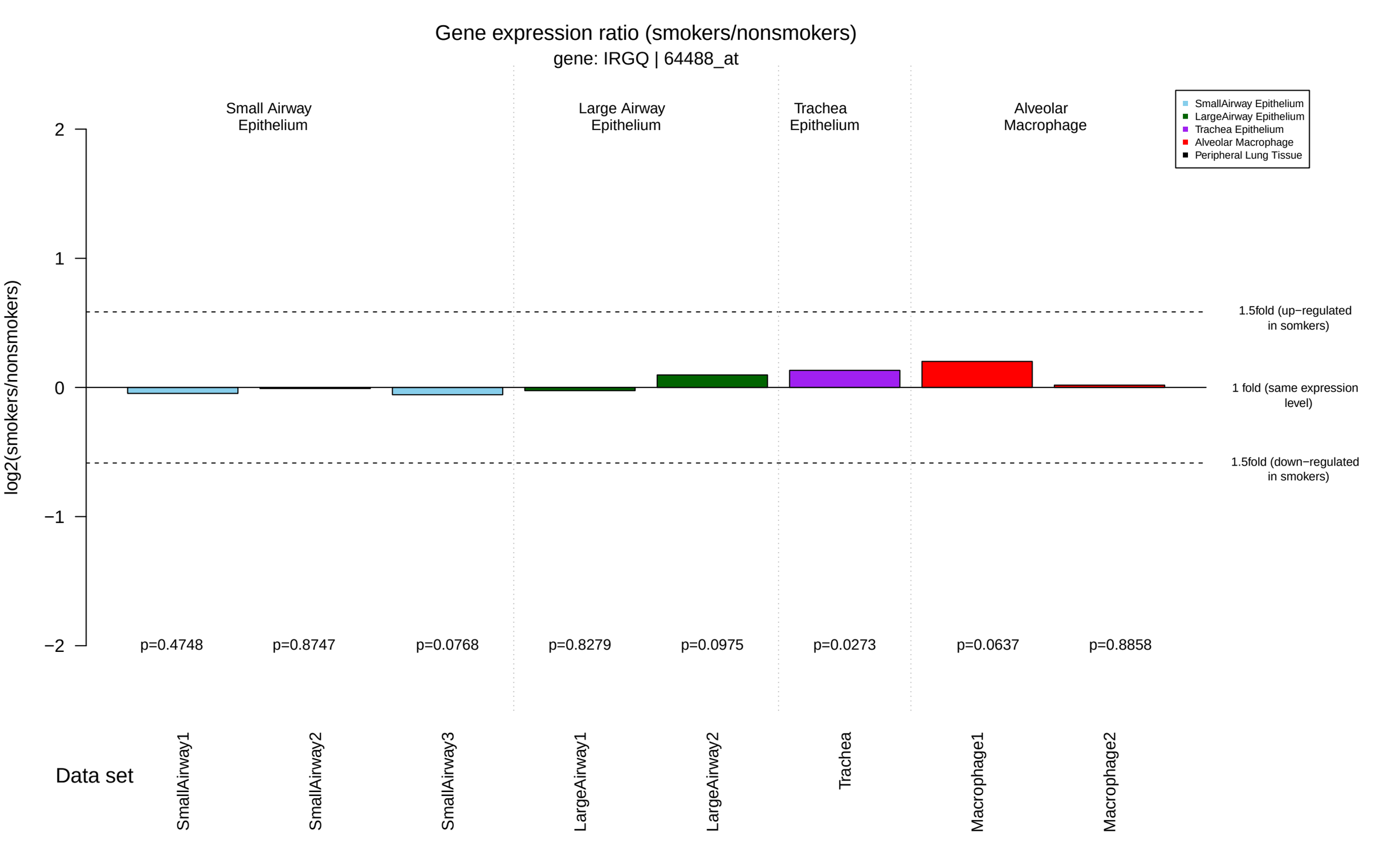
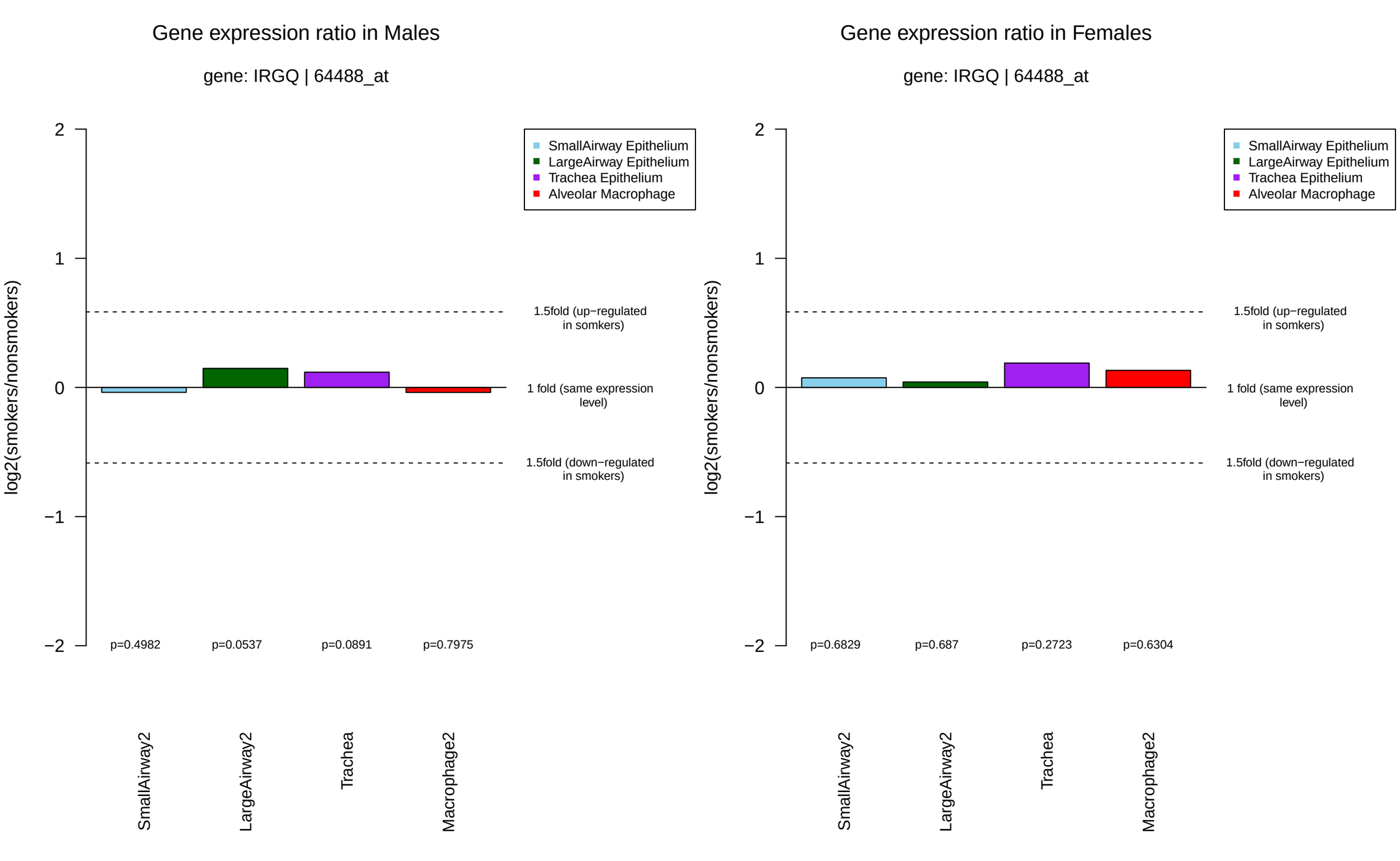
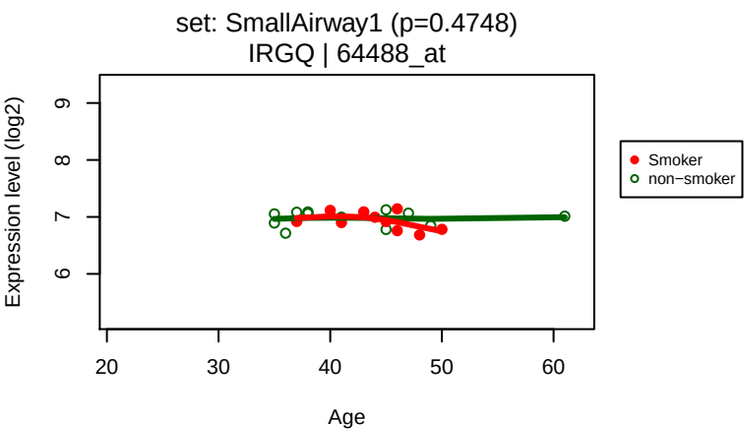
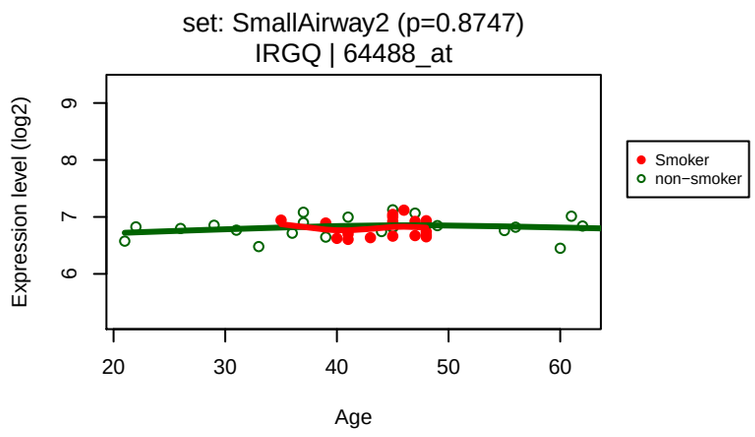
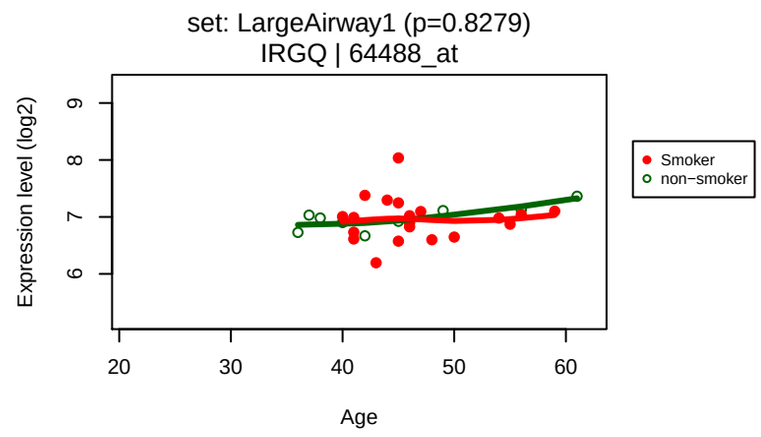

Gene expression of smokers and nonsmokers
gene: IRGQ | 64488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4748
p=0.8747
p=0.0768
p=0.8279
p=0.0975
p=0.0273
p=0.0637
p=0.8858
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

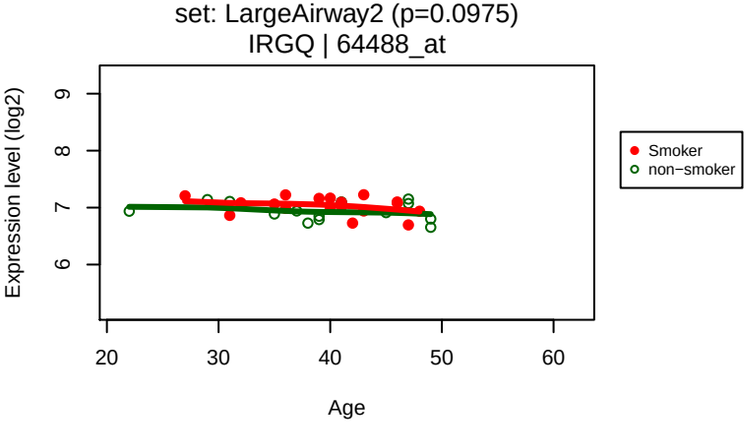

Gene expression ratio (smokers/nonsmokers)
gene: IRGQ | 64488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4748
p=0.8747
p=0.0768
p=0.8279
p=0.0975
p=0.0273
p=0.0637
p=0.8858
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

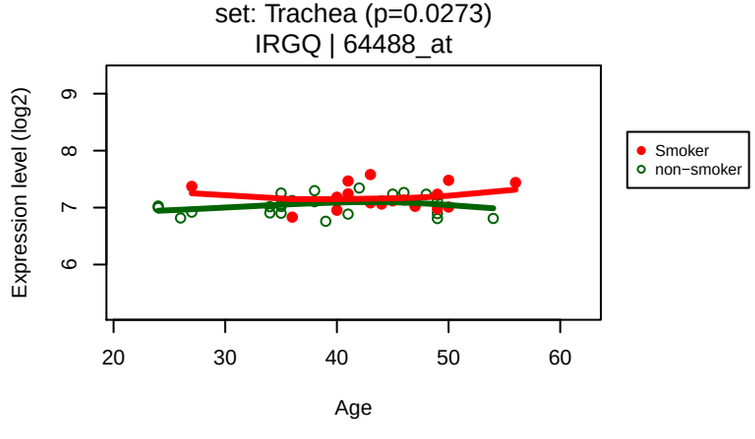

Gene expression ratio in Males
gene: IRGQ | 64488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4982
p=0.0537
p=0.0891
p=0.7975
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRGQ | 64488_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6829
p=0.687
p=0.2723
p=0.6304
SmallAirway2
LargeAirway2
Trachea
Macrophage2

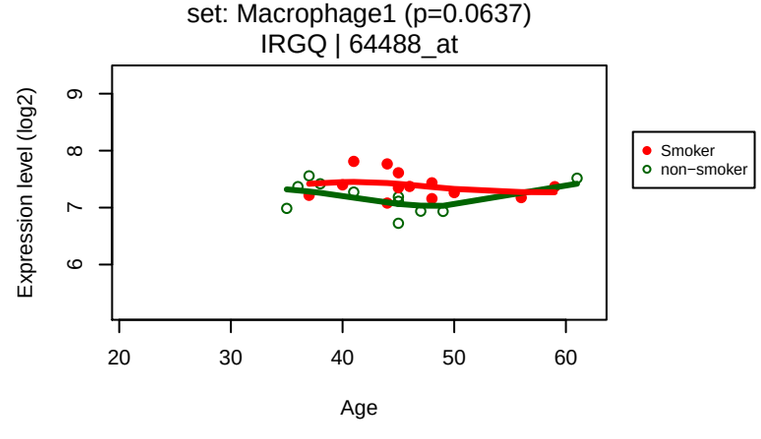

set: SmallAirway1 (p=0.4748)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker

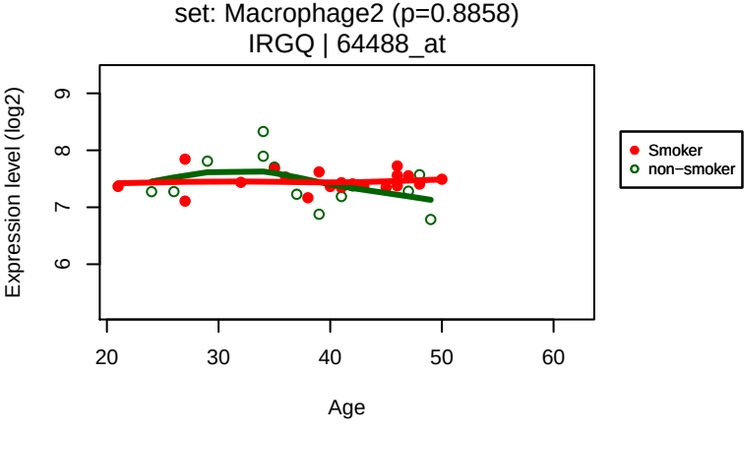

set: SmallAirway2 (p=0.8747)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker

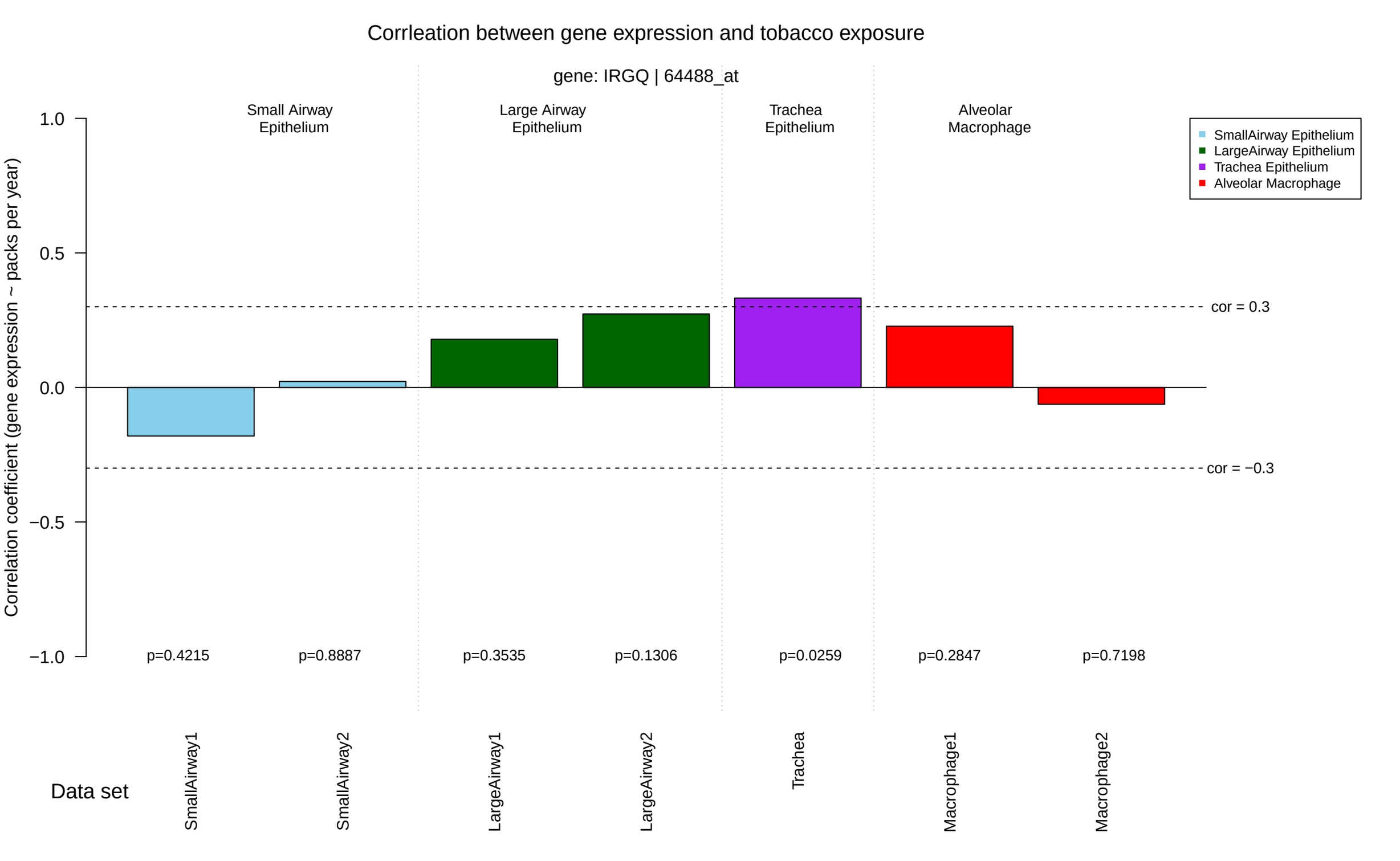

set: LargeAirway1 (p=0.8279)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0975)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0273)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0637)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8858)
IRGQ | 64488_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRGQ | 64488_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.4215
p=0.8887
p=0.3535
p=0.1306
p=0.0259
p=0.2847
p=0.7198
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2